Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/30/2014
PI Name:	Cowings, Patricia S. Ph.D.		
Project Title:	Examining Individual Differences in Temporal Profiles of Cardiovascular Responses to Head Down Tilt During Fluid Loading		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Patricia.S.Cowings@nasa.gov	**Fax:**	FY 650-604-1484
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5724
Organization Name:	NASA Ames Research Center		
PI Address 1:	Human Systems Integration Division (TH)		
PI Address 2:	Mail Stop 262-2		
PI Web Page:	http://humansystems.arc.nasa.gov/		
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/27/2012	**End Date:**	09/30/2013
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Toscano, William (NASA Ames Research Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Background: Susceptibility of healthy astronauts to orthostatic hypotension and presyncope is exacerbated upon return from spaceflight. Up to 64% of astronauts experience post-flight orthostatic intolerance. A promising countermeasure for post-flight orthostatic intolerance is fluid loading by giving crew salt tablets and water prior to re-entry. The primary goal of the current study was to determine the optimal time window prior to re-entry when individual crewpersons would initiate fluid loading. Methods: Sixteen participants were given two, 6-hour exposures, with and without fluid loading, to Head-Down Tilt (HDT) to simulate the effects of microgravity. Pre- and post-HDT stand tests of orthostatic tolerance were given. Physiological responses recorded included measures of heart rate, blood pressure, peripheral blood volume, total peripheral resistance, and impedance cardiography. Echocardiography measures of stroke volume and cardiac output were also recorded. Results: Data were analyzed with three-way repeated measures ANOVA (gender, condition		

	(no fluid, fluid) and time). Only the condition x time interaction was significant for mean arterial pressure. Post-hoc multiple comparisons tests revealed significant increases in mean arterial pressure occurred between hours 1 and 3 of HDT after fluid loading ($p<0.05$). Discussion: These findings indicate that the optimal time for crew to begin fluid loading is within 1 to 3 hours prior to re-entry. Non-significant trends of multiple cardiovascular responses showed similar time profiles. The large amount of individual variability suggests that fluid loading alone may be an inadequate countermeasure for all crewmembers. Further research is needed on possible adjunct methods of tailoring countermeasures for individuals.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	By simultaneously monitoring multiple cardiovascular responses to head-down tilt and fluid loading, new insights were found that may explain individual differences in orthostatic tolerance and the effectiveness of this countermeasure. Continuous measures of peripheral circulation and impedance cardiography may help to define more effective treatments for patients on Earth suffering from postural orthostatic intolerance or hypotension.
Task Progress:	Background: Susceptibility of healthy astronauts to orthostatic hypotension and presyncope is exacerbated upon return from spaceflight. Up to 64% of astronauts experience post-flight orthostatic intolerance. A promising countermeasure for post-flight orthostatic intolerance is fluid loading by giving crew salt tablets and water prior to re-entry. The primary goal of the current study was to determine the optimal time window prior to re-entry when individual crewpersons would initiate fluid loading. Methods: Sixteen participants were given two, 6-hour exposures, with and without fluid loading (condition), to Head-Down Tilt (HDT) to simulate the effects of microgravity. Pre- and post-HDT stand tests of orthostatic tolerance were given. Physiological responses recorded included measures of heart rate, blood pressure, peripheral blood volume, total peripheral resistance, and impedance cardiography. Echocardiography measures of stroke volume and cardiac output were also recorded. Results: Physiological data (30-minute means) from HDT tests were analyzed with repeated measures 3-way ANOVA: gender x condition x time (HDT hours 1 to 6). Gender was not significant for any of the measures. Condition was significant for finger pulse volume and mean arterial pressure, and the time was significant for all measures. The condition x time interaction was significant for mean arterial pressure, skin conductance, thoracic fluid volume, and total peripheral resistance. Tukey-Kramer tests revealed significantly higher mean arterial pressures at 60, 90, 120, and 150 minutes following fluid loading HDT when compared to no-fluid HDT. None of the other variables showed significant differences at these time-points. Echocardiography measures (30-second means) for gender and condition were not significant for either stroke volume or cardiac output, while the time was significant only for cardiac output. However, there were no significant two or three-way interactions for either measure. Only one participant experienced syncope and another subject was pre-syncopal during stand tests following no-fluid loading HDT. Analyses of physiological data from stand tests (pre vs. post HDT) showed no significant gender or condition effects for any of the measures, while the main effect of stand tests was significant for all measures except skin temperature, skin conductance, and stroke volume. The gender x condition interaction was significant for mean arterial pressure, cardiac output measured with impedance cardiography, and total peripheral resistance with men generally higher, although not significantly, than women after fluid loading. The condition x stand interaction was only significant for mean arterial pressure. Tukey-Kramer tests for mean arterial pressure of all subjects revealed significant increases from pre- to post- stand tests for both two conditions; however, there was no difference between the post-test following fluid loading and no fluid loading. Conclusion: These findings indicate that the optimal time for crew to begin fluid loading is within 1 to 3 hours prior to re-entry. Non-significant trends of multiple cardiovascular responses showed similar time profiles. The large amount of individual variability suggests that fluid loading alone may be an inadequate countermeasure for all crewmembers. Further research is needed on possible adjunct methods of tailoring countermeasures for individuals.
Bibliography Type:	Description: (Last Updated: 11/08/2023)
Abstracts for Journals and Proceedings	Cowings PS, Toscano WB, Kanis D, Saicheur T, Ravikumar A, Gebreyesus F. "Examining Individual Differences in the Temporal Profiles of Cardiovascular Responses to Head-Down Tilt with and without Fluid Loading: Narrowing a Gap." Cardiovascular Session 1. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3127.pdf , Feb-2014
Articles in Peer-reviewed Journals	Cowings P, Toscano W, Kanis D, Saicheur T, Ravikumar A, Gebreyesus F. "Fluid loading effects on temporal profiles of cardiovascular responses to head-down tilt." Aerosp Med Human Perform. 2015 Feb;86(2):88-96. PMID: 25946732 , Feb-2015